CLINICAL PROGRAMMING BOOTCAMP • MODULE 01

Introduction to CDISC & SDTM Foundations

How clinical trial data becomes a standardized, submission-ready dataset — from raw case report forms to a regulatory package.

CDISC · CDASH · SDTM · ADaM · Controlled Terminology · Define-XML

WHY STANDARDS EXIST

Before standards: every study spoke a different language

Same idea, different names
One study calls a subject SUBJID, another PATIENT, another PT_NO — for the exact same thing.

Different structures
Adverse events stored one-row-per-event here, one-column-per-event there.

Different codes
Sex recorded as M/F, 1/2, Male/Female, or Homme/Femme across studies.

Re-learning every time
A reviewer opening a new submission had to decode the data from scratch.

The reviewer's nightmare

Study A `SUBJID | SEX=1 | AE in wide format`

Study B `PATIENT | SEX=M | AE one row per event`

Study C `PT_NO | SEX=Male | AE free-text notes`

Every submission = a new puzzle for the FDA reviewer.

WHY STANDARDS EXIST

Standards turn re-work into re-use

1

SHARED STRUCTURE

Learn the data model once, apply it to every study and every sponsor.

100%

REQUIRED BY FDA & PMDA

Standardized study data is mandatory for regulatory submissions.

Faster

REVIEW & REUSE

Reviewers, tools, and analysis programs all know what to expect.

A data standard is a shared agreement on structure, names, and codes — so data means the same thing everywhere.

THE ORGANIZATION

What is CDISC?

Clinical Data Interchange Standards Consortium

A global, non-profit organization
founded in 1997. It brings together industry, regulators, and technology providers to develop and maintain open data standards for clinical research.

Its standards are vendor-neutral, freely published, and used worldwide across the drug-development lifecycle.

- ✓ **Common structure**
  How datasets and variables are organized
- ✓ **Common naming**
  What each variable is called
- ✓ **Common codelists**
  The allowed values (Controlled Terminology)
- ✓ **Common metadata**
  How the data is described to reviewers

THE STANDARDS FAMILY

One family, one lifecycle

COLLECTION

CDASH

Standard for how data is captured on case report forms (CRFs).

TABULATION

SDTM

Organizes collected data into standard domains for submission.

ANALYSIS

ADaM

Analysis-ready datasets derived from SDTM, traceable to it.

Controlled Terminology — the standardized codelists (allowed values) used across all of the above.

Define-XML — the machine-readable “data dictionary” that describes every dataset and variable to the regulator.

THE CLINICAL DATA JOURNEY

From patient visit to regulatory submission

01

COLLECTION

CDASH

Data captured at sites on CRFs / EDC systems.

02

TABULATION

SDTM

Raw data mapped into standard submission domains.

03

ANALYSIS

ADaM

Analysis datasets derived; tables & figures produced.

04

SUBMISSION

eCTD

Package sent to regulators (FDA / PMDA) for review.

SDTM is the bridge: it standardizes collected data so analysis and regulatory review can build on a common foundation.

THE STANDARD IN FOCUS

What is SDTM — and why it matters

Study Data Tabulation Model

SDTM defines a standard structure for organizing and formatting the data collected in a clinical trial — so that the *tabulations* (the observed, as-collected data) are consistent across studies and sponsors.

Think of SDTM as a set of labeled, standard-shaped boxes. Whatever the study, the same kind of data goes in the same box, with the same label.

Required for submission
FDA and Japan's PMDA mandate SDTM-formatted study data.

Consistency
Every study is organized the same way — easy to combine and compare.

Enables automation
Validation tools and review software rely on the predictable structure.

Traceability
A clear starting point that analysis (ADaM) datasets trace back to.

FOUNDATIONAL DOCUMENTS

SDTM Model vs. SDTMIG

SDTM (the Model)

The general framework

- Defines the underlying concepts and rules
- Introduces the General Observation Classes
- Lists the variable roles (Identifier, Topic, Timing, Qualifier)
- Standard-wide — not tied to any one domain

SDTMIG (the Guide)

Implementation Guide — how to actually do it

- Turns the model into concrete, buildable domains
- Specifies exact variables for DM, AE, VS, LB, ...
- Gives assumptions, examples, and business rules
- This is the book programmers open every day

Model = the rules of the game. SDTMIG = the detailed playbook. We use SDTMIG v3.3 throughout this bootcamp.

THE CORE IDEA

Everything is an observation

Almost all trial data fits one of three General Observation Classes. The class determines the shape of the domain.

INTERVENTIONS

Something given to or done to the subject

Treatments, therapies, procedures — investigational or not.

Examples: CM (concomitant meds), EX (exposure)

EVENTS

Something that happened to the subject

Occurrences and milestones during the trial.

Examples: AE (adverse events), MH (medical history), DS (disposition)

FINDINGS

Something measured or assessed

The result of an observation or test.

Examples: VS (vital signs), LB (labs), EG (ECG)

HOW DOMAINS ARE GROUPED

The domain categories

1 Special Purpose

Unique structure, don't follow the 3 classes.

DM · CO · SE · SV

2 Interventions

Treatments & procedures given to the subject.

CM · EX · EC · PR · SU

3 Events

Things that happened to the subject.

AE · MH · DS · CE

4 Findings

Measurements & assessments.

VS · LB · EG · PE · QS

5 Trial Design

Describe the study itself, not subjects.

TA · TE · TS · TV

6 Relationships

Link records across domains.

RELREC · SUPPQUAL

REFERENCE

Common domains cheat sheet

Domain	Name	Class / Category	What it holds
DM	Demographics	Special Purpose	One row per subject: age, sex, race, arm
AE	Adverse Events	Events	Untoward medical occurrences
CM	Concomitant Meds	Interventions	Other medications the subject took
EX	Exposure	Interventions	Study treatment actually administered
LB	Laboratory	Findings	Lab test results (chemistry, hematology...)
VS	Vital Signs	Findings	BP, pulse, temperature, weight, height
MH	Medical History	Events	Relevant prior/ongoing conditions
DS	Disposition	Events	Subject's status / completion / withdrawal
EG	ECG	Findings	Electrocardiogram results
PE	Physical Exam	Findings	Physical examination findings

These 10 cover the backbone of most studies — you'll build several of them by hand in this bootcamp.

ANATOMY OF A DATASET

One row = one observation (DM example)

In Demographics (DM), the observation is the subject — so there is exactly one row per subject.

STUDYID	DOMAIN	USUBJID	SUBJID	AGE	AGEU	SEX	RACE	ARMCD	ARM	COUNTRY
ABC-01	DM	ABC-01-01-001	001	54	YEARS	F	WHITE	A	Drug A	USA
ABC-01	DM	ABC-01-01-002	002	60	YEARS	M	ASIAN	P	Placebo	JPN
ABC-01	DM	ABC-01-01-003	003	46	YEARS	F	BLACK OR AFRICAN AMERICAN	A	Drug A	USA

Rows = observations

Each row is one subject. Three subjects → three rows. No subject appears twice in DM.

Columns = variables

Each column is one standardized variable with a fixed name (STUDYID, AGE, SEX...).

Values follow rules

SEX = F/M and RACE values come from Controlled Terminology — not free text.

STRUCTURE OF A DOMAIN

Four variable roles

IDENTIFIER

Who & which record

Identifies the study, domain, subject, and record.

`STUDYID · DOMAIN · USUBJID · AESEQ`

TOPIC

What the observation is about

The focus of the row — the thing observed.

`AETERM · LBTESTCD · CMTRT`

TIMING

When it happened

Dates, times, and study day of the observation.

`AESTDTC · VSDY · VISIT`

QUALIFIER

Describes / adds detail

Everything that characterizes the topic value.

`AESEV · LBORRES · VSORRESU`

Every SDTM variable plays exactly one role. Most variables are Qualifiers.

WHICH VARIABLES MUST YOU INCLUDE?

Core: Required / Expected / Permissible

Req	**Required**	Must always be present, and must be populated — never left blank.	*e.g.* STUDYID, DOMAIN, USUBJID, --SEQ, and the topic variable.
Exp	**Expected**	Must be present, but a value may be missing if genuinely not collected.	*e.g.* AETERM's timing (AESTDTC), VSORRES, etc.
Perm	**Permissible**	Include only if collected / relevant to the study. Optional.	*e.g.* Extra qualifiers like AESER, VSPOS.

The SDTMIG marks every variable Req / Exp / Perm. Start every domain by including all Required and Expected variables.

THE FOUR YOU'LL USE CONSTANTLY

Key identifier variables

STUDYID

`ABC-01`

Study Identifier

The unique id of the study. Same value on every row of every domain.

DOMAIN

`DM`

Domain Abbreviation

The 2-letter domain code. Constant within a dataset.

USUBJID

`ABC-01-01-001`

Unique Subject Identifier

Unique per subject across the WHOLE study. Links a subject across all domains.

--SEQ

`AESEQ = 1, 2, 3 …`

Sequence Number

Numbers a subject's records within a domain so each row is uniquely identifiable.

USUBJID is the thread that stitches a subject's data together across DM, AE, VS, LB and every other domain.

--TESTCD / --TEST: the vertical structure

Findings domains are tall & narrow: one row per test result. The test is named in --TESTCD/--TEST; the result goes in --ORRES.

USUBJID	VSSEQ	VSTESTCD	VSTEST	VSORRES	VSORRESU	VSSTRESN	VISIT
ABC-01-01-001	1	**SYSBP**	Systolic Blood Pressure	120	mmHg	120	BASELINE
ABC-01-01-001	2	**DIABP**	Diastolic Blood Pressure	78	mmHg	78	BASELINE
ABC-01-01-001	3	**PULSE**	Pulse Rate	66	beats/min	66	BASELINE
ABC-01-01-001	4	**TEMP**	Temperature	36.8	C	36.8	BASELINE
ABC-01-01-001	5	**SYSBP**	Systolic Blood Pressure	118	mmHg	118	WEEK 2

Why “tall”? Adding a new test (e.g., HEIGHT) means adding rows, not new columns. The structure never changes.

ORRES vs STRESN. VSORRES is the original result as collected; VSSTRESN is the standardized numeric result for analysis.

Controlled Terminology (CT)

The allowed-values dictionary

CT is the set of standardized codelists that define the permitted values for coded SDTM variables.

Maintained by CDISC with NCI-EVS and updated quarterly. It removes ambiguity: one concept, one value, everywhere.

Examples of coded variables: SEX, RACE, AESEV, AESER, VSTESTCD, units.

Free text → Controlled value

Free text		Controlled value	Variable
“Female”, “F”, “2”	→	**F**	SEX
“severe”, “Sev”, “3”	→	**SEVERE**	AESEV
“mild”, “Mild”, “1”	→	**MILD**	AESEV
“bpm”, “beats per min”	→	**beats/min**	VSORRESU

THE SUBMISSION PACKAGE

Define-XML: the map that ships with the data

What Define-XML is

A machine-readable metadata file (the “data definition”) that describes every dataset, variable, codelist, and derivation in the submission.

It is the reviewer's map to your data — arguably as important as the datasets themselves.

A typical SDTM submission package

1. **SDTM datasets**
   The domains themselves (SAS XPORT .xpt files)
2. **Define-XML**
   Metadata describing all datasets & variables
3. **cSDRG**
   Study Data Reviewer's Guide — narrative context
4. **annotated CRF**
   Blank CRF marked with the SDTM variable names

Raw adverse-event data (as collected)

This is how an adverse event might look in the raw EDC extract — human-friendly, but not standardized.

Subject	AE Term	Start Date	End Date	Severity	Serious?	Outcome (code)
001	bad headache	15/03/2024	16/03/2024	moderate	No	1
002	mild dizziness	10-Mar-2024	12-Mar-2024	Mild	N	1

What's “wrong” for SDTM?

- ! Inconsistent dates: DD/MM/YYYY vs DD-Mon-YYYY — SDTM needs ISO 8601 (YYYY-MM-DD).
- ! Free-text terms: “bad headache” must be coded (e.g., via MedDRA) and split into reported vs. dictionary terms.
- ! Severity as free text: must map to Controlled Terminology (MILD / MODERATE / SEVERE).
- ! Serious as No / N: must become standard N / Y in AESER.
- ! Outcome is a CODE (1-5): must be decoded to its CT term, e.g. 1 → RECOVERED/RESOLVED.
- ! Subject “001” isn't unique study-wide — subject 001 exists at BOTH sites. Needs a full USUBJID.

The same data, mapped to SDTM AE

STUDYID	DOMAIN	USUBJID	AESEQ	AETERM	AEDECOD	AESTDTC	AEENDTC	AESEV	AESER
ABC-01	AE	ABC-01-01-001	1	bad headache	**Headache**	2024-03-15	2024-03-16	**MODERATE**	N
ABC-01	AE	ABC-01-01-002	1	mild dizziness	**Dizziness**	2024-03-10	2024-03-12	**MILD**	N

Dates → ISO 8601

15/03/2024 → 2024-03-15

Term coded (MedDRA)

bad headache → AEDECOD Headache

Severity → CT

moderate → MODERATE

Outcome code → CT

1 → RECOVERED/RESOLVED

Serious → CT

No / N → N

Subject → USUBJID

site 01 + 001 → ABC-01-01-001

Record numbered

AESEQ assigned per subject

AETERM keeps the verbatim term; AEDECOD holds the coded term. Both matter — SDTM preserves what was collected and adds standardization.

LEARN FROM THESE EARLY

Common beginner pitfalls

Inventing variable names
Use the SDTMIG's exact names. No custom columns in standard domains — use SUPPQUAL for extras.

Wrong date format
All SDTM dates are ISO 8601 character (YYYY-MM-DD). Never store dates as free text or numbers.

Dropping Expected variables
Expected columns stay in the dataset even if some values are blank.

Ignoring Controlled Terminology
Coded values must match the CT codelist exactly, including spelling and case.

Non-unique USUBJID
USUBJID must be unique study-wide and identical across all domains for one subject.

Confusing ORRES and STRESN
ORRES = original as collected; STRESN/STRESC = standardized for analysis.

QUICK REFERENCE

Glossary of acronyms

CDISC	Clinical Data Interchange Standards Consortium
CDASH	Clinical Data Acquisition Standards Harmonization
SDTM	Study Data Tabulation Model
SDTMIG	SDTM Implementation Guide
ADaM	Analysis Data Model
CT	Controlled Terminology
CRF / eCRF	Case Report Form (electronic)
EDC	Electronic Data Capture
MedDRA	Medical Dictionary for Regulatory Activities
USUBJID	Unique Subject Identifier
FDA / PMDA	US / Japan regulatory agencies
cSDRG	clinical Study Data Reviewer's Guide

WHAT'S NEXT IN THE BOOTCAMP

From concepts to building domains

1. **Tooling** SAS and R environments & language basics — set up to code in both.
2. **Importing data** Reading raw EDC/CRF CSV files into SAS and R.
3. **Building domains** Map raw data into DM, AE, CM, EX, VS & LB — step by step, in both languages.
4. **Deriving & checking** Study day (--DY), --SEQ, Controlled Terminology, and Pinnacle 21-style validation.
5. **Capstone** Map a small mock study to several SDTM domains, end to end.

Every hands-on exercise is provided in both SAS and R — you'll build the same domain twice and see how each language does it.